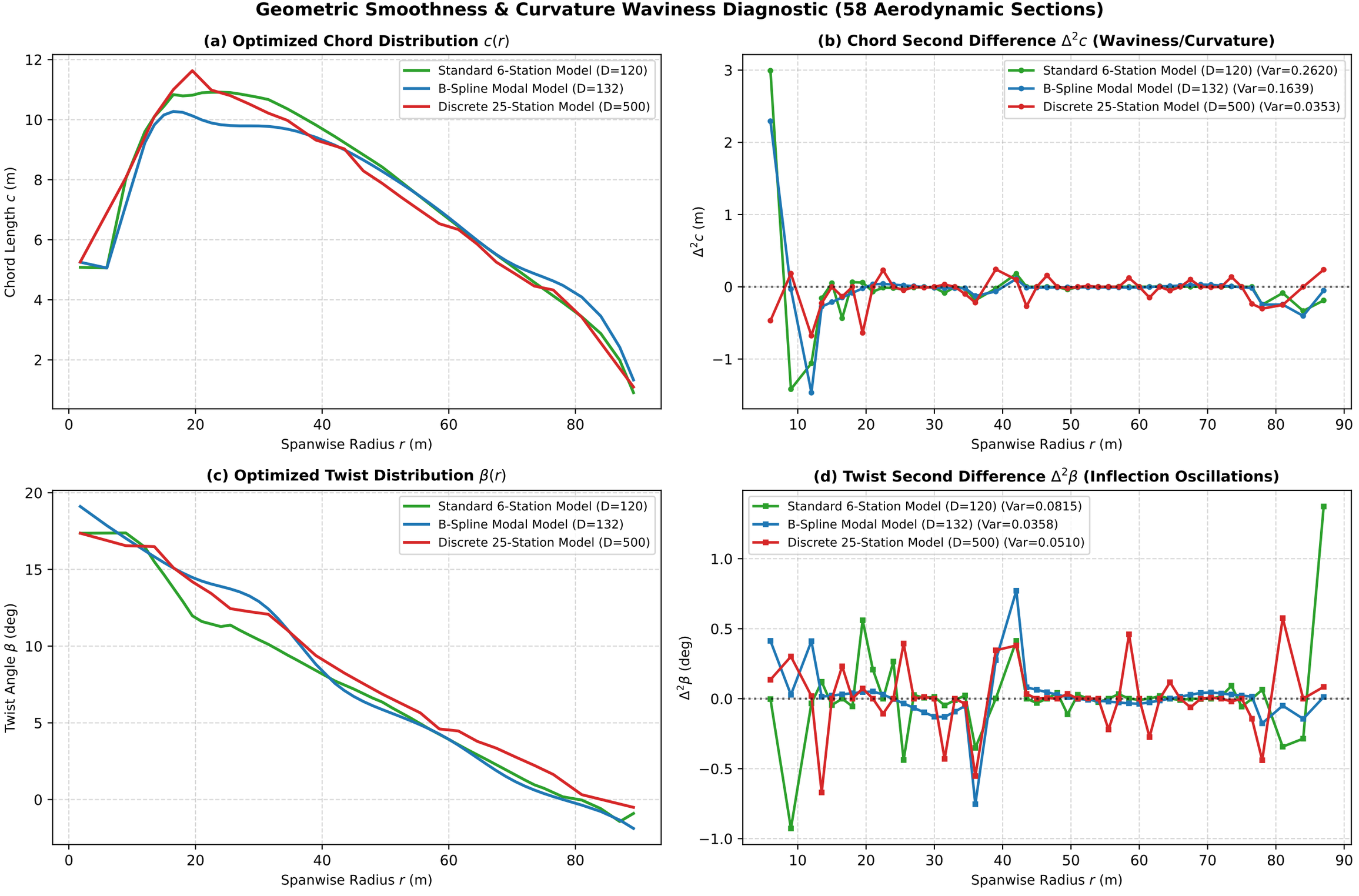
Geometric Smoothness & Curvature Waviness Diagnostic (58 Aerodynamic Sections)
(a) Optimized Chord Distribution $c(r)$
Chord Length c (m)
Spanwise Radius r (m)
Standard 6-Station Model (D=120)
B-Spline Modal Model (D=132)
Discrete 25-Station Model (D=500)
(b) Chord Second Difference $\Delta^2 c$ (Waviness/Curvature)
$\Delta^2 c$ (m)
Spanwise Radius r (m)
Standard 6-Station Model (D=120) (Var=0.2620)
B-Spline Modal Model (D=132) (Var=0.1639)
Discrete 25-Station Model (D=500) (Var=0.0353)
(c) Optimized Twist Distribution $\beta(r)$
Twist Angle β (deg)
Spanwise Radius r (m)
Standard 6-Station Model (D=120)
B-Spline Modal Model (D=132)
Discrete 25-Station Model (D=500)
(d) Twist Second Difference $\Delta^2 \beta$ (Inflection Oscillations)
$\Delta^2 \beta$ (deg)
Spanwise Radius r (m)
Standard 6-Station Model (D=120) (Var=0.0815)
B-Spline Modal Model (D=132) (Var=0.0358)
Discrete 25-Station Model (D=500) (Var=0.0510)